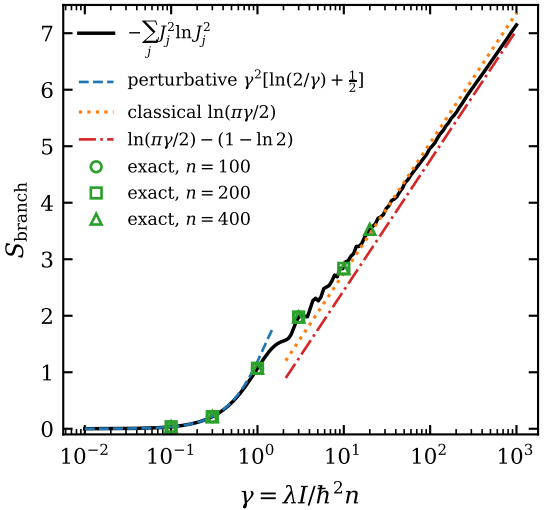
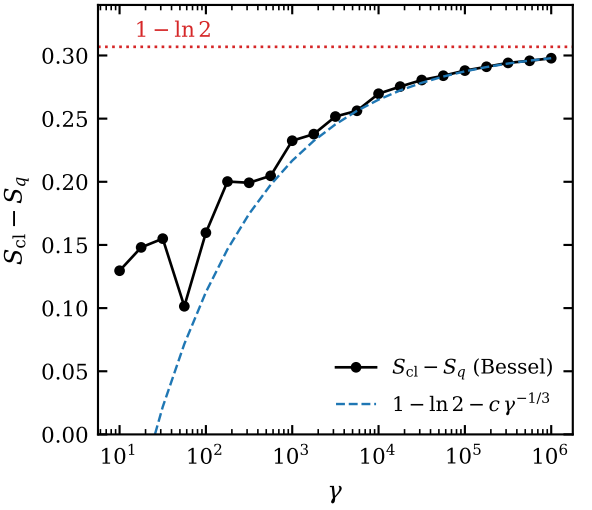

(a) deep rotation: universal crossover
$-\sum_j J_j^2 \ln J_j^2$
perturbative $\gamma^2[\ln(2/\gamma)+\frac{1}{2}]$
classical $\ln(\pi\gamma/2)$
$\ln(\pi\gamma/2)-(1-\ln 2)$
exact, $n=100$
exact, $n=200$
exact, $n=400$
$S_{\rm branch}$
$\gamma=\lambda I/\hbar^2 n$


(b) approach to $1-\ln 2$
$1-\ln 2$
$S_{\rm cl}-S_q$
$S_{\rm cl}-S_q$ (Bessel)
$1-\ln 2-c\,\gamma^{-1/3}$
γ